

# Cloud CRM System - Data Model & Entity Relationship Design

Confidential - Internal Use Only

Version 1.0

## 1. Core Entities

Entity: Lead - Fields: lead\_id (UUID PK), first\_name (VARCHAR 100), last\_name (VARCHAR 100), email (VARCHAR 255 UNIQUE), phone (VARCHAR 20 NULL), company\_name (VARCHAR 200), title (VARCHAR 150 NULL), source (ENUM: web\_form, email, api, manual, referral, event), status (ENUM: new, contacted, qualified, nurture, converted, disqualified), score (SMALLINT 0-100 DEFAULT 0), assigned\_to (UUID FK users NULL), converted\_contact\_id (UUID FK contacts NULL), created\_at (TIMESTAMP), updated\_at (TIMESTAMP).

Entity: Contact - Fields: contact\_id (UUID PK), first\_name (VARCHAR 100), last\_name (VARCHAR 100), email (VARCHAR 255), phone (VARCHAR 20 NULL), mobile (VARCHAR 20 NULL), title (VARCHAR 150 NULL), company\_id (UUID FK companies), lifecycle\_stage (ENUM: subscriber, lead, mql, sql, customer, evangelist), owner\_id (UUID FK users), clearbit\_data (JSONB NULL), custom\_fields (JSONB DEFAULT {}), created\_at (TIMESTAMP), updated\_at (TIMESTAMP).

Entity: Company - Fields: company\_id (UUID PK), name (VARCHAR 300), domain (VARCHAR 255 UNIQUE NULL), industry (VARCHAR 100 NULL), employee\_count (INTEGER NULL), annual\_revenue (BIGINT NULL), parent\_company\_id (UUID FK companies NULL SELF-REF), clearbit\_data (JSONB NULL), created\_at (TIMESTAMP), updated\_at (TIMESTAMP).

Entity: Deal - Fields: deal\_id (UUID PK), name (VARCHAR 300), company\_id (UUID FK companies), primary\_contact\_id (UUID FK contacts), pipeline\_id (UUID FK pipelines), stage\_id (UUID FK pipeline\_stages), value (DECIMAL 15,2), currency (CHAR 3 DEFAULT 'USD'), probability (SMALLINT 0-100), expected\_close\_date (DATE NULL), actual\_close\_date (DATE NULL), won (BOOLEAN NULL), loss\_reason (VARCHAR 200 NULL), owner\_id (UUID FK users), created\_at (TIMESTAMP), updated\_at (TIMESTAMP).

Entity: Activity - Fields: activity\_id (UUID PK), type (ENUM: call, email, meeting, note, task), subject (VARCHAR 500), body (TEXT NULL), contact\_id (UUID FK contacts NULL), deal\_id (UUID FK deals NULL), owner\_id (UUID FK users), duration\_minutes (SMALLINT NULL), scheduled\_at (TIMESTAMPTZ NULL), completed\_at (TIMESTAMPTZ NULL), outcome (VARCHAR 100 NULL), created\_at (TIMESTAMP).

Entity: Pipeline - Fields: pipeline\_id (UUID PK), name (VARCHAR 100), team\_id (UUID FK teams NULL), is\_default (BOOLEAN DEFAULT false), created\_at (TIMESTAMP). PipelineStage - Fields: stage\_id (UUID PK), pipeline\_id (UUID FK pipelines), name (VARCHAR 100), position (SMALLINT), probability (SMALLINT 0-100), is\_won (BOOLEAN DEFAULT false), is\_lost (BOOLEAN DEFAULT false).

Entity: EmailThread - Fields: thread\_id (UUID PK), contact\_id (UUID FK contacts), deal\_id (UUID FK deals NULL), subject (VARCHAR 500), message\_count (INTEGER DEFAULT 1), provider (ENUM: gmail, outlook), provider\_thread\_id (VARCHAR 255), last\_message\_at (TIMESTAMPTZ), is\_tracked (BOOLEAN DEFAULT true), opens (INTEGER DEFAULT 0), clicks (INTEGER DEFAULT 0).

## 2. Relationships

Company 1:N Contact - A company has many associated contacts.

Company 1:N Deal - Deals are associated with a company.

Contact N:N Deal - Contacts can be involved in multiple deals (junction table: deal\_contacts).

Deal N:1 Pipeline - Each deal belongs to one pipeline.

Deal N:1 PipelineStage - Each deal is in one stage at a time.

Contact 1:N Activity - Activities are logged against contacts.

Deal 1:N Activity - Activities can also be linked to deals.

Contact 1:N EmailThread - Email conversations belong to contacts.

User 1:N Lead - Sales reps own leads.

User 1:N Deal - Sales reps own deals.

Company 0:1 Company (self-referential) - Parent-child company hierarchy.

### 3. Search & Indexing

REQ-CRM-074: Elasticsearch indices for: leads (name, company, email), contacts (name, email, company), deals (name, company), activities (subject, body). Full-text search with fuzzy matching and highlighting.

REQ-CRM-075: Real-time index updates via CDC (Change Data Capture) using Debezium from PostgreSQL WAL. Index lag < 2 seconds.

REQ-CRM-076: Search suggestions: type-ahead with 3-character minimum, returns top 5 results per entity type (leads, contacts, deals) with entity-type icon.

### 4. Data Partitioning & Tenant Isolation

REQ-CRM-077: Multi-tenant architecture using tenant\_id column on all tables. Row-level security (RLS) policies enforce tenant isolation at database level.

REQ-CRM-078: Activity and EmailThread tables partitioned by created\_at quarter. Retention: 3 years in hot storage, 7 years in cold archive.

REQ-CRM-079: Materialized views for dashboards: daily\_pipeline\_snapshot, weekly\_activity\_summary, monthly\_revenue\_forecast. Refreshed every 10 minutes.